

Instruction:

1) The ProgMods tab can be copied (as required) to create multiple ProgMods tabs, according to the programming module in the study
2) Update the "Title" of the ProgMods to be descriptive (e.g. 5A Baseline Characteristics)

3) List the newly added ProgMods tab(s) in Section C of the Cover tab

4) Delete any examples in grey italic text that are not relevant/applicable to the study once the programming specification has been finalized

Title10.LOT1_BASE_END

Section A: Defining Medication Available Period (MAP) medications									
Time Period	Variable	Label	Values	Definition	Code Lists Group	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)	
PD	LOT1_END_DT_TEMP	Temporary end date for LOT1 base period		The LOT will continue until the earliest of any of the following: (1) Discontinuation of all agents in the regimen (with or without switch to a new agent) - the LOT end date is based on the run-out date, defined as the last date that any MAP medication in that LOT is considered to be available; (2) Unplanned SCTs - if an autologous SCT occurs > 180 days following a previous autologous SCT, the LOT will end the day before the latter SCT; any allogeneic SCTs are considered a new LOT; (3) SCTs not followed by maintenance within 180 days - the last day of the LOT is the date of the SCT; (4) Death; (5) Health plan disenrollment; (6) End of the study period; (7) End of maintenance regimen. Note: Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT (Rule 1 - this is an exception, not an ending event).		LOT1_BASE_END_DT derived from LOT1_BASE_DISCON_DT (max MAP_END_DT from T_MEDICAL_PROC_CD/NDC and T_RX_NDC/OT/induction meds), LOT1_BASE_1ST_ADO_MED_DT (first MAP_START_DT from T_MEDICAL/T_RX for non-induction meds minus 1), LOT1_TX_ENDDATE (earliest LOT-ending SCT from T_MEDICAL procedure codes minus 1), and OBS_END_DT (LEAST of study_end from CLM.DEATH_DT from T_DOD YMOD, disenrollment from T_MEMBER_CONF_ENROLLMENT_ELIGEND).			
	LOT1_END_REASON_TEMP	Temporary LOT1 base period end reason		The LOT ends for one of these reasons: (Rule 2) Discontinuation of all agents in the regimen, with or without switch to a new agent; (Rule 2) If the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration/dispense date of the new agent; (Rule 3) Unplanned SCTs - if an autologous SCT occurs > 180 days following a previous autologous SCT, or any allogeneic SCT; (Rule 4) SCTs not followed by maintenance within 180 days; (Rule 5) Death; (Rule 6) Health plan disenrollment; (Rule 7) End of the study period; (Rule 8) End of maintenance regimen.		End reason derived from matching the LOT1 end date against dates from: T_MEDICAL/T_RX (discontinuation via MAP_END_DT), T_MEDICAL/T_RX (new med via MAP_START_DT), T_DOD (death via YMOD), T_MEDICAL (SCT via procedure codes in T_MED_PROCDURE), ENDDATE from T_MEMBER_CONF_ENROLLMENT (disenrollment) and study_end config.		JM06Apr 2026	
	LOT1_HSCT_AUTO_SING_FLAG	Flag: base period had single valid autologous HSCT		Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. SCT is operationalized as >1 procedure code for an SCT. A single autologous SCT is one that is not part of a valid tandem pair. Single autologous SCTs and tandem SCTs are not unplanned SCTs and are considered a continuation of the line of therapy.		Autologous HSCT identified from T_MEDICAL procedure codes (PROC_CD, BILL_PROC_CD) matching SCT codelist entries for AUTO type. Claims grouped by 14-day windows from T_MEDICAL_FST_DT. Gap validation (60 days) and tandem detection (180 days) applied to grouped dates.		JM06Apr 2026	
	LOT1_HSCT_AUTO_TAND_FLAG	Flag: base period had tandem valid autologous HSCT		Patients will be classified as having a planned autologous tandem SCT if they receive a second autologous SCT more than 60 days and less than 180 days after the first planned autologous SCT within the same line of therapy (LOT). If the interval between the two autologous SCTs falls outside this window, the sequence will be considered unplanned. An autologous SCT followed by an allogeneic SCT will not be considered planned, because allogeneic SCTs typically follow failed autologous SCTs.		Tandem detection uses T_MEDICAL procedure codes for both AUTO events. The 180-day window is applied to grouped AUTO dates (after 14-day windowing and 60-day gap validation). ALLO events between AUTO_DT_1 and AUTO_DT_2 checked via T_MEDICAL.			
PD	LOT1_TX_AUTO_MAX_DT	Latest date of patient's valid HSCT		This variable captures the date of the second valid tandem autologous (AUTO) SCT - where a tandem is two AUTOs >60 and <180 days apart - or, if only a single AUTO occurred, the date of that single AUTO.		Derived from T_MEDICAL procedure code dates for autologous HSCT, after 14-day windowing and 60-day gap validation.			
PD	LOT1_TX_AUTO_FLAG	Flag: patient had any valid autologous HSCT		Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. This binary variable indicates whether the patient received any valid autologous transplant, regardless of single or tandem.		Derived flag from T_MEDICAL procedure code analysis for autologous HSCT events.		JM06Apr 2026	
PD	LOT1_BASE_MEDS	LOT1 induction medications		The LOT1 regimen is defined as all multiple myeloma (MM) oncology therapies received on and within 60 days of the LOT1 start date, with the first therapy received considered the induction therapy. This variable therefore captures the full set of oncology agents administered during this 60-day window, excluding non-oncology agents such as corticosteroids. Switches between a biologic reference product and its biosimilar(s), or between biosimilars of the same reference product, are permitted and do not constitute a change in regimen or advance the line of therapy. (Madduri, 2021).		Medications identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MMMA_CODELIST (Tab 41). Medication abbreviations from CL_MMMA_ROLLUP (Tab 40). Claims filtered by MAP_START_DT within induction window from LOT1_START_DT.			
PD	LOT1_BASEMAINT_START	Start date of maintenance period		A maintenance regimen is a period of 120 days or longer during which only a valid maintenance therapy is available. Maintenance is not required to be prescribed continuously for the duration of the 120 day period. Furthermore, the maintenance therapy must have been a part of the LOT's initial regimen.		Maintenance medications identified from T_MEDICAL/T_RX claims matched to CL_MMMA_ROLLUP medications with NONMAINTENANCE=1 or valid DUAL MAINTENANCE WITH combinations. Start date is the first MAP_START_DT of the maintenance medication after LOT1_BASE_END_DT.			

	A	B	C	D	E	F	G	H	I	J
	LOT1_BASEMAINT_TYP	Maintenance medication abbreviation	valid maintenance therapies per NCCN Guidelines include lenalidomide monotherapy, bortezomib monotherapy, daratumumab monotherapy, azacitidine monotherapy, thalidomide monotherapy, bortezomib/lenalidomide dual, carfilzomib/lenalidomide dual, and daratumumab/lenalidomide dual. Once a maintenance regimen has begun, it continues until discontinuation, presence of an SCT, night censoring (i.e. end of follow-up), or the addition of any non-maintenance PMT therapy automatically causes the maintenance regimen to end.				Medication abbreviation from CL_MMA_ROLLUP (Tab 40) MED_ABBR field, identifying the maintenance drug from T_RX (INDC) or T_MEDICAL (PROC_CD/INDC) claims.			
	LOT1_BASEMAINT_END	End date of maintenance period (temp)	Flag for maintenance medication (MED) (BCORT/DABA/XMZ/LE NAC/CAR)				Maintenance end date derived from MAP_END_DT for maintenance medications in T_MEDICAL/RX, compared against other end events from the same source tables.			
	LOT1_BASEMAINT_MED (MED)	Flag for maintenance medication (MED) (BCORT/DABA/XMZ/LE NAC/CAR)	valid maintenance therapies per NCCN Guidelines (Kumar, 2020) include mono-maintenance (lenalidomide, bortezomib, daratumumab, azacitidine, thalidomide) and dual-maintenance (bortezomib/lenalidomide, carfilzomib/lenalidomide, daratumumab/lenalidomide). Any of the maintenance medications listed above will be considered maintenance medications for all patients, regardless of transplant status.				Individual medication flags derived from T_RX (INDC) and T_MEDICAL (PROC_CD/INDC) claims matched to CL_MMA_CODELIST (Tab 41) and CL_MMA_ROLLUP (Tab 40) for specific medication abbreviations.			
	LOT1_BASEMAINT_END_REASON	Maintenance period end reason	The LOT will end after a maintenance regimen ends due to: unplanned SCTs, death, health plan disenrollment, and end of study period. The addition of any non-maintenance PMT therapy automatically causes the maintenance regimen to end.				End reason derived from the same Optum CDH source tables as LOT1_END_REASON, JEPP: T_MEDICAL/RX for medication events, T_DOD to death, T_MEMBER_COUNT_ENROLLMENT for disenrollment.			
	LOT1_BASE_END_DT	Final LOT1 base period end date	The LOT will continue until the earliest of any of the following: (Rules 2, 8 are ending events; Rule 1 from the protocol defines permissible substitutions that do NOT end the LOT). Per Rule 2: The LOT end date is based on the run-out date, defined as the last date that any PM medication in that LOT is considered to be available. If the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration/disperse date of the new agent.				LOT1_BASE_END_DT derived from priority evaluation of dates from T_MEDICAL/RX (MAP_END_DT for discontinuation, MAP_START_DT for new med), T_MEDICAL procedure codes (SCT events), T_DOD (YVNDOD for death), T_MEMBER_COUNT_ENROLLMENT (EUGEND for disenrollment), and study_end config parameter.			
	LOT1_BASE_END_REASON	Final LOT1 base period end reason	LOT ends for one of the following reasons (Rules 2-8 from the protocol): (2) Discontinuation of all agents; (3) Unplanned SCTs; (4) SCTs not followed by maintenance within 180 days; (5) Death; (6) Health plan disenrollment; (7) End of study period; (8) End of maintenance regimen, Rule 1 (permissible substitutions - biologic reference product with biosimilars) does not advance the LOT and is not an ending event.				End reason derived from evaluation of dates across T_MEDICAL/RX (medication events), T_MEDICAL (SCT procedure codes), T_DOD (death), and T_MEMBER_COUNT_ENROLLMENT (disenrollment). Priority logic applied analytically.			
	FIRST_ALLO_DT	Date of first allogeneic HSCT during LOT1	Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. Rule 3: Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other PMT therapies included in that LOT.				ALLO SCT identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=ALLO. Date from FST_DT in T_MEDICAL. Sequential ordering applied to identify first ALLO event.			
	FIRST_CART_DT	Date of first CAR-T therapy during LOT1	CAR-T: CAR-T cellular therapy infusions are classified as their own LOT. Therapies (including supportive agents, e.g., corticosteroids) given within 30 days of CAR-T cellular therapy are consolidated as part of the CAR-T LOT.				CAR-T identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=CART. Date from FST_DT in T_MEDICAL.			
	LOT_DISCON_GAP_DAYS	Discontinuation gap threshold (days)	Discontinuation of the LOT1 regimen occurs when all PM oncology agents in the line of therapy are stopped, if the regimen is discontinued and no new agent is initiated, the LOT end date is defined as the run-out date of the last administered agent. Sensitivity analyses will assess the robustness of the line of therapy rules by varying key algorithm parameters				Gap calculated as datediff(OBS_END_DT, RAW_DISCON_DT) where OBS_END_DT is derived from T_MEMBER_COUNT_ENROLLMENT (EUGEND), T_DOD (YVNDOD), and study_end config. RAW_DISCON_DT is max(MAP_END_DT) from T_MEDICAL/RX for induction medications.			
	MEDICAL_DAY_SUPPLY	Assumed days supply for medical claims	The days supply on medical claims is always missing and therefore a 28-day supply is assumed. When pharmacy claims are missing days supply, or when days supply values are anomalous, a 28-day supply is assumed.				Applied to T_MEDICAL claims where DAYS_SUPPLY is NULL. For RX pharmacy claims, DAYS_SUPPLY is typically populated but when missing/anomalous, 28 days is substituted. The assumed supply extends MAP_END_DT = FST_DT (or FILL_DT) + days_supply - 1.			
	LOT1_BASE_LENGTH	LOT1 base period duration in days	The protocol does not define LOT1_BASE_LENGTH as a named variable. Duration is implied by Protocol Section 5.1.1 which defines LOT start date (LOT1 begins on the date of the first fill for an PM therapy following a patient's index date) and LOT end date (earliest of Rules 1-8 in the protocol).				Duration computed from dates already derived from T_MEDICAL/RX (LOT1_START_DT from first MAP_START_DT) and LOT1_BASE_END_DT (priority-based end date).			
	LOT1_SCT_TO_MAINT_180D_RULE	SCT not followed by maintenance within 180 days - LOT end rule	SCTs not followed by maintenance: If an SCT is not followed by a maintenance regimen within 180 days, then the last day of the LOT is the date of the SCT. Following an autologous SCT (within 180 days) or in the case of a tandem autologous SCT within 180 days of the second SCT, the expectation after an SCT is for a maintenance period and thus instead of the usual 120 days, we check for a valid 30 day (minimum) maintenance period.				Would require checking T_MEDICAL/RX claims for maintenance medications (PMO/MO/MAINTENANCE=1) or valid DUAL/MAINTENANCE WITH in CL_MMA_ROLLUP within 180 days after the SCT date (LOT1_1ST_SCT_DT or LOT1_TX_AUTO_MAX_DT).			
	PERMISSIBLE_SUBS_EFFECT	Permissible substitution rules affecting LOT1 end	Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered within the same line, which is named according to the drug given for the longest duration.				Substitutions tracked via CL_MMA_ROLLUP (Tab 40) which maps medication abbreviations. The permissible subs reference table pairs original med with substitute_med. When a new medication claim appears in T_MEDICAL/RX, it is checked against permissible_subs before being classified as a new medication addition.			
	LOT1_TX_ENDDATE_REASON	SCT end reason type	The type of stem cell transplant (SCT) determines how LOT1 ends. In the case of an autologous SCT, an				SCT type determined from T_MEDICAL procedure codes matched against			

	A	B	C	D	E	F	G	H	I
	LOT1_TX_ENDDATE_REASON (AUTO/ALLO/CART)	SCT end/reason type (AUTO/ALLO/CART)			The type of stem cell transplant (SCT) determines how LOT1 ends. In the case of an autologous SCT, an unplanned transplant ends the current line of therapy the day before the SCT date. For an allogeneic SCT, the current LOT ends the day before the allogeneic SCT date, and a new LOT begins from the date of the transplant. CAR-T cellular therapy infusions are classified as their own line of therapy, with the preceding LOT ending the day before the CAR-T infusion date; however, if a new oncology agent is introduced within 45 days of the CAR-T infusion, that agent is incorporated into the CAR-T LOT rather than initiating a new line of therapy.		SCT type determined from T_MEDICAL procedure codes matched against CL_SCT CODELIST (Tab 4a) which classifies procedures as AUTO, ALLO, or CART. Priority is earliest LOT-ending date wins.		
U_PD	LOT1_TX_AUTO_DT_1/ LOT1_TX_AUTO_DT_2	Dates of first and second valid autologous HSCT			"Planned Tandem SCT: Patients will be classified as having a planned tandem SCT if they receive at least two autologous SCTs >90 to <180 days apart." SCT is operationalized as >1 procedure code for an SCT. "Tandem SCTs are two SCTs >90 to <180 days apart. These are considered planned and a continuation of the line of therapy. Codes for SCTs are in the programming specification document."		AUTO HSCT dates derived from T_MEDICAL procedure codes (PROC_CD, BILL_PROC_CD) matched against CL_SCT_CODELIST where SCT_TYPE= AUTO. FST_DT from T_MEDICAL is the raw claim date. 14-day windowing groups claims; 60-day gap validation separates distinct events.		JY06Apr/2026
U_PD	LOT1_BASE_DISCON_DT	LOT1 base period discontinuation date			"Discontinuation of the regimen occurs when all MM agents in the LOT are stopped, either without initiation of a new MM agent (all agents have reached their run-out dates and no further MM administration/dispenses are observed), or with initiation of a new MM agent that was not present in the induction regimen." The LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available.		max(MAP_END_DT) derived from T_MEDICAL (FST_DT + medical_day_supply_1) and TX_FILL_DT + DAYS_SUPPLY_1 for induction medications matched via CL_MMA_CODELIST. Gap validated against OBS_END_DT from T_MEMBER_CONT_ENROLLMENT/ DOD.		JY06Apr/2026
U_PD	LOT1_BASE_1ST_ADD_MED_DT /LOT1_BASE_1ST_ADD_MED	Date and name of first non-induction medication added			"If the current LOT is immediately interrupted by a new agent (the new agent starts on or before the run-out date of the current regimen), the LOT end date is the day before the first administration/dispense date of the new agent." Per Rule 1: permissible substitutions (biologic reference product with biosimilars) do not advance the LOT.		First non-induction medication identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MMA_CODELIST (Tab 41) where the medication is NOT in base_meds (induction + permissible subs). MAP_START_DT from MMA algorithm.		JY06Apr/2026
U_PD	INDUCTION_WINDOW_DAYS	Induction regimen identification window (days)			"The LOT1 regimen includes all MM therapies received within 60 days (Paddur, 2021) on and following the LOT1 start date." For LOT2-LOT5, the LOT regimen includes all MM therapies received within 30 days on and following the LOT start date.		Applied to MAP_START_DT from T_MEDICAL/ RX claims. Medications with MAP_START_DT within the window are classified as induction via CL_MMA_CODELIST/CL_MMA_ROLLUP matching.		JY06Apr/2026
U_PD	ALLO_ALWAYS_ENDS_LOT	Allogeneic SCT always ends current LOT (protocol rule)			'Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT.' 'An autologous SCT followed by an allogeneic SCT will not be considered planned because an allogeneic SCT typically follows failed autologous SCTs.' Rule 3: Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs.' 'Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.'		ALLO SCT identified from T_MEDICAL procedure codes matched against CL_SCT_CODELIST where SCT_TYPE= ALLO. Date from FST_DT in T_MEDICAL.		JY06Apr/2026
U_PD	CART_30D_CONSOLIDATION	CAR-T 30-day supportive agent consolidation rule			CAR-T cellular therapy infusions are classified as their own line of therapy. Oncology therapy administered within 45 days of the CAR-T infusion are consolidated as part of the CAR-T LOT.		Would require checking T_MEDICAL/ RX claims within 30 days of FIRST_CART_DT and reclassifying them from LOT1 to the CAR-T LOT. Affects MAP_END_DT for induction meds and MAP_START_DT for potential new meds.		
U_PD	NAMING_DISCREPANCY_SCT_V S_HSCT	Variable naming: spec uses HSCT prefix; code uses SCT prefix			The protocol uses SCT (Stem Cell Transplant) terminology consistently throughout. Stem cell transplant (SCT). The original program spec uses HSCT (Hematopoietic Stem Cell Transplant) prefix for variable names. Both abbreviations refer to the same medical concept.		N/A - naming convention only.		JY06Apr/2026
U_PD					If LOT1 contains any of the following, in combination with at least one other agent then flag 'yes': - Lenalidomide - Bortezomib - Daratumumab - Ixazomib - Thalidomide - Bortezomib/lenalidomide - Carfilzomib/lenalidomide - Daratumumab/lenalidomide else =no		For this LOT algorithm, the team has decided not to define maintenance therapy as its own LOT but rather to flag the inclusion of LOT1 regimens that include potential valid maintenance therapeutic/therapy combinations.		
U_PD	contains_mnk_req	1 = yes 0 = no					Definition of a maintenance period: The LOT's initial regimen transitions into a maintenance regimen, such that non-maintenance medications present during the initial regimen are discontinued/leaving only maintenance medications for x days. It is thus imperative to include another agent, other than the drug(s) that transitions into a mnk regimen to anchor the start of that mnk regimen.		